

An Empirical Study of Book-to-Market Portfolios in the U.S. Market (1982–2021)

BAFI508 Data-driven Investments

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Introduction

Motivation

- The book-to-market (B/M) ratio has historically predicted stock returns, with value stocks outperforming growth stocks.
- Widely used in empirical asset pricing and quantitative investing.
- A modern re-evaluation with updated data and methods can reveal insights into its current relevance and possible enhancements.

Literature

- Fama & French (1992, 1993): B/M in 3-factor model
- Fama & French (2015): Added profitability & investment
- Hou et al. (2020): B/M effect weakened post-2000
- Lakonishok et al. (1994): Behavioral explanation for value premium

What We Hope to Find

- Replication: Perform B/M portfolio sorts using CRSP & Compustat, comparing results with Ken French's dataset.
- Performance Analysis: Evaluate long-short returns, summary statistics, and factor regressions (CAPM, Fama-French).
- Extension: Double sort on B/M and operating profitability to test for enhanced return predictability.
- Outcome: Assess the robustness of the B/M effect in recent data and its variation across industries and firm sizes.

Methodology

1. Data cleaning and Preparation

- CRSP Market Data:
 - Filtered for common stocks (SHRCD 10, 11) on NYSE, AMEX, and NASDAQ (EXCHCD 1, 2, 3).
 - Market Cap: $|\text{Price}| \times \text{Shares Outstanding}$
- Compustat Book Equity Data:
 - Book Equity = Shareholders' Equity + Deferred Taxes - Preferred Stock.

2. Construct Book-to-Market Ratio

- Merged Compustat BE (from $t-1$) with CRSP Market Cap (from June of year t)
- B/M ratio = Book Equity / Market Cap

3. Sort stocks into 5 portfolios based on B/M

- Used B/M ratio in June to assign stocks into 5 quantiles

- Quintile 0 = Lowest B/M (growth stocks)
- Quintile 4 = Highest B/M (value stocks)

4. Track monthly returns (July t to June $t+1$)

For each portfolio, collected monthly returns from CRSP

5. Computed equal-weighted returns for each quintile

Run CAPM & Fama-French 3-Factor regressions

Example:

Time Period	Data Used	Source
End of 2018($t-1$)	Book Equity	Compustat
Jun 2019 (t)	Market Cap	CRSP
Jun 2019 (t)	B/M = BE/ME	Merge
July 2019 - Jun 2020 (t to $t+1$)	Monthly Returns	CRSP

Results and Interpretation

Average Returns Table

Metric	Qnt 1	Qnt 2	Qnt 3	Qnt 4	Qnt 5	Long-Short
Mean	0.007811	0.011513	0.013506	0.014309	0.019392	0.011581
t-stat	2.303541	3.925740	4.883473	5.409108	6.063301	6.090364
p-value	0.021715	0.000100	0.000001	0.000000	0.000000	0.000000
SD	0.071131	0.061517	0.058012	0.055491	0.067088	0.039886
Skewness	-0.314275	-0.365000	-0.492325	-0.521835	0.233643	0.568097
Kurtosis	2.700754	3.038127	3.176564	3.974902	4.399952	2.469230

- **Excess Return:** Returns exceeding the expected return based on a given model.
- **Average Returns Table:**
 - All Quintiles (Qnt 1 to Qnt 5) and Long-Short Portfolio show positive mean returns.
 - Statistically significant t-values across all Quintiles .
 - Low p-values (except Qnt 1, marginal at 0.0217).
- **Conclusion:** These portfolios generate positive returns that may be considered excess returns, especially if not fully explained by risk factors.

CAPM Results Table

Metric	Qnt 1	Qnt 2	Qnt 3	Qnt 4	Qnt 5	Long-Short
Alpha	-0.005161	-0.000435	0.002175	0.003528	0.008116	0.010463
Alpha t	-2.717221	-0.293669	1.507797	2.409296	3.681565	5.563423
Alpha p	0.006844	0.769150	0.132327	0.016395	0.000261	0.000000
Beta Mkt	1.314736	1.182048	1.102242	1.031030	1.095109	-0.220001
Beta Mkt t	31.714633	36.598446	35.014017	32.257022	22.760150	-5.359741
Beta Mkt p	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
R²	0.696638	0.753579	0.736776	0.703757	0.541853	0.061550

Fama-French 3-Factor Results Table

Metric	Qnt 1	Qnt 2	Qnt 3	Qnt 4	Qnt 5	Long-Short
Alpha	-0.004826	-0.000911	0.001032	0.001956	0.005749	0.007760
Alpha t	-4.115582	-0.980024	1.154626	2.134447	3.647266	6.080734
Alpha p	0.000046	0.327618	0.248876	0.033363	0.000297	0.000000
Beta Mkt	1.142385	1.065808	1.015188	0.967835	1.027380	-0.116502
Beta Mkt t	43.417261	51.108739	50.611068	47.066135	29.046822	-4.068320
Beta Mkt p	0.000000	0.000000	0.000000	0.000000	0.000000	0.000056
Beta SMB	0.927794	0.809986	0.812814	0.784935	1.020183	0.099148
Beta SMB t	23.183245	25.536791	26.641738	25.096534	18.963483	2.276344
Beta SMB p	0.000000	0.000000	0.000000	0.000000	0.000000	0.023309
Beta HML	-0.312673	0.002326	0.240358	0.400003	0.635754	0.947224
Beta HML t	-8.162770	0.076623	8.231035	13.361865	12.346774	22.721191
Beta HML p	0.000000	0.938959	0.000000	0.000000	0.000000	0.000000
R²	0.886012	0.904180	0.900319	0.885635	0.769081	0.573971

CAPM vs. Fama-French 3-Factor Analysis

- Alpha (Excess Return)
 - Portfolios 3, 4, and 5 all have positive alphas, suggesting value portfolios earn higher returns.
 - Implies higher B/M portfolios have significantly positive excess returns
 - Implies a value premium: value stocks (high B/M) outperform growth stocks (low B/M)
- Alpha t-stat
 - Quintile 1: statistically significant negative alpha
 - Quintile 4: statistically significant positive alpha
 - Quintile 5: strong, significant positive alpha
- Long-Short Portfolio
 - Significant alpha values vs near-zero alpha
- R²
 - Portfolios 1–5 have high R² (mostly > 0.88), showing the FF3 model fits well
 - Long-Short R² = 0.57, lower fit suggests alpha is not captured by FF3, implying potential mispricing or missing factors.

Portfolio Performance: Cumulative and Long-Short Returns

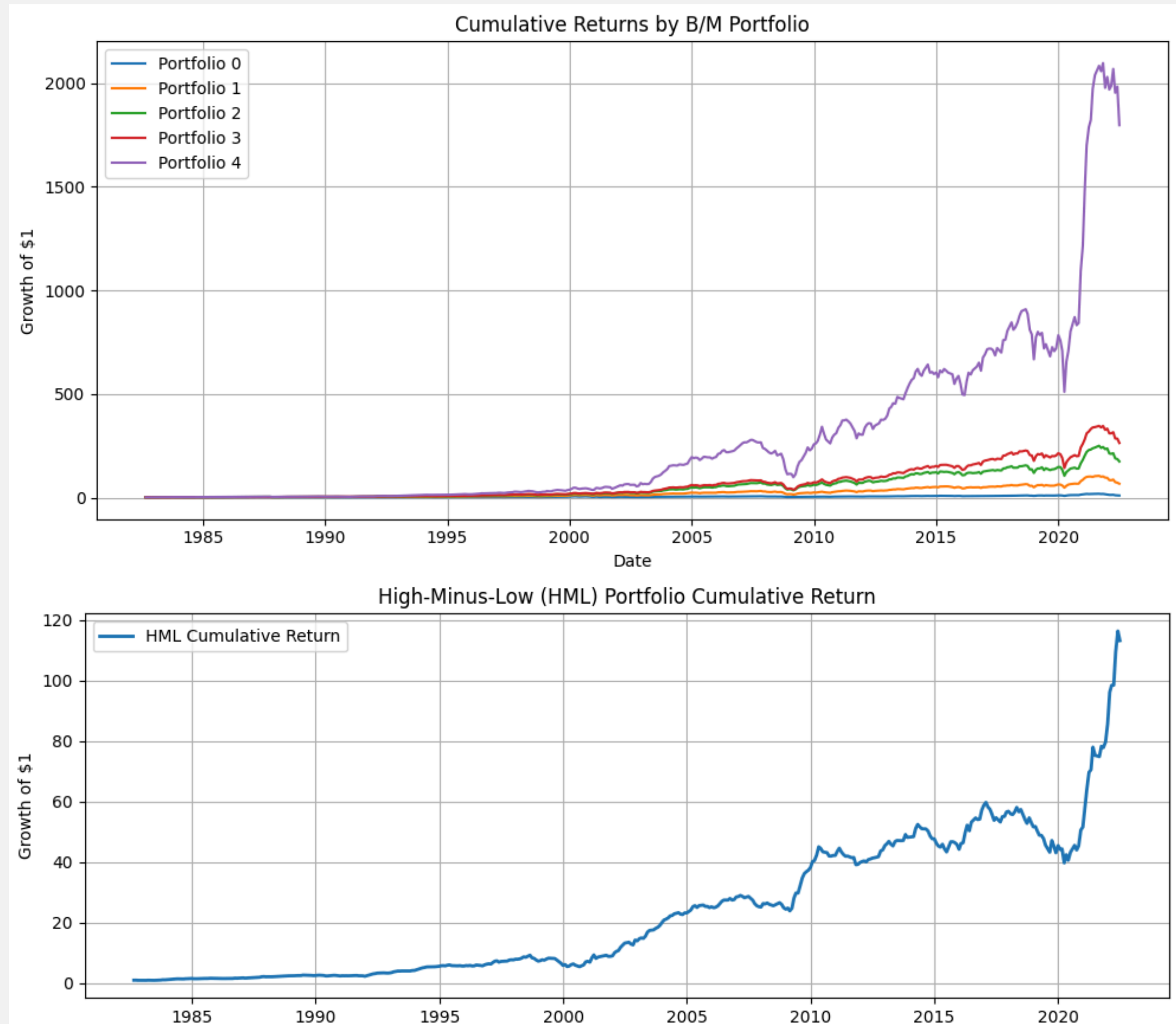
Cumulative Returns by B/M Quintile:

- Portfolios are formed based on Book-to-Market (B/M) quintiles from 1982 to 2021.
- Portfolio 4 (High B/M, “Value” stocks) significantly outperforms others.
- Clear upward trend shows value stocks delivered higher long-term returns.

High-Minus-Low (HML) Portfolio:

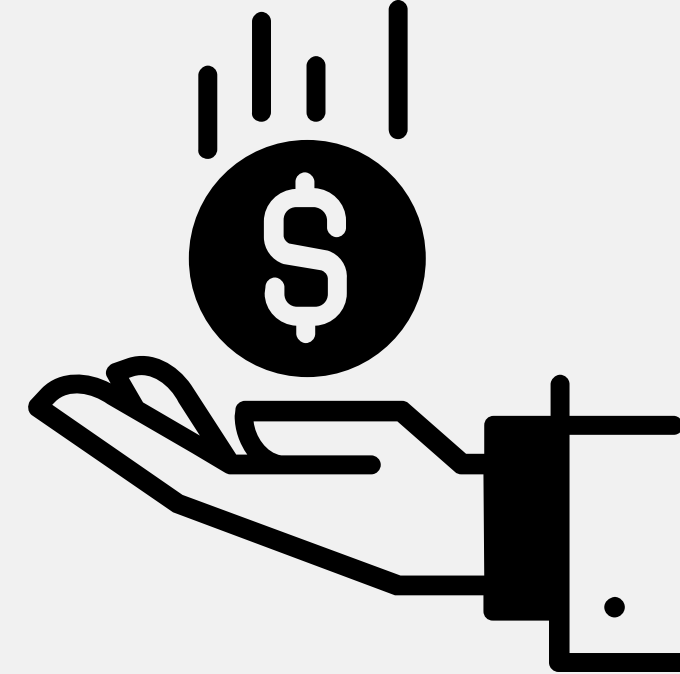
- Constructed as Portfolio 4 minus Portfolio 0 (High B/M – Low B/M).
- HML portfolio captures the value premium.
- Annualized return: 13.77%, reflecting strong historical performance of value strategies.

The HML strategy consistently beats growth stocks and provides robust long-term returns.



Conclusion

- **Excess Return**
 - Portfolios 4 and 5 have the excess return (statistically significant)
- **Compare findings from the literature**
 - The high B/M portfolio has higher expected returns.
 - The Long-Short portfolio shows overall positive returns.
 - The three-factor model significantly improves explanatory power
- **Compare findings from the data on Ken French's website**
 - NYSE, AMEX, and NASDAQ vs only NYSE data.
 - equal-weighted vs value-weighted.



Limitation

- The univariate in the FF3-Factor model may not fully capture the complexity of the factor.
- More advanced multi-factor models or additional variables may be considered.



Proposed Extension

- Fama-French 5-Factor Model Expansion
- Add RMW (profitability) and CMA (investment) factors
- Add momentum as a sixth factor

The background is a solid green color with a subtle texture. It is decorated with various white line-art icons related to finance and business. These include stacks of money, individual coins, dollar signs, a piggy bank, a hand holding coins, a bar chart, a calculator, a briefcase, and a target with a dollar sign. The icons are scattered around the perimeter of the image, framing the central text.

Thank you!